



Universal Computing Technologies

in association with upskill Campus (USC) & The IoT Academy

Industrial Internship Report

PROJECT 4 — AGRICULTURE

Prediction of Agriculture Crop Production in India

Dataset: Government of India | data.gov.in | 2001–2014

Field	Details
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Organisation	UCT — Universal Computing Technologies
Internship Programme	upskill Campus × UCT Machine Learning Internship
Project	Project 4 — Agriculture Crop Production Prediction
Date	May 2026
Duration	6 Weeks
Domain	Machine Learning / Data Science / Agriculture

Executive Summary

This report documents a 6-week Machine Learning internship at UCT (Universal Computing Technologies) through upskill Campus, focused on building an end-to-end crop yield prediction system using publicly available Government of India data.

India's agriculture sector, contributing ~17–18% of GDP and employing 54% of the workforce, suffers from chronic yield unpredictability. This project addresses that gap by developing a supervised ML model trained on 2001–2014 data from data.gov.in to predict crop yield (Quintals/Hectare) for seven major Indian crops across multiple states.

Six regression algorithms were evaluated. The Random Forest Regressor delivered the best test-set performance with an R^2 of 0.9377 and MAE of 31.45 Q/Ha. A Rice production forecast for 2015–2019 was also generated using linear trend extrapolation. The best model was serialised with joblib for production deployment.

Metric	Value
Best Model	Random Forest Regressor
Test R^2 (Accuracy)	0.9377 (93.77% variance explained)
Test MAE	31.45 Quintals / Hectare
Test RMSE	75.04 Quintals / Hectare
Cross-Validation R^2 (Gradient Boosting)	0.915 (Best generalisation)
Crops Covered	ARHAR, COTTON, GROUNDNUT, MAIZE, PADDY, SUNFLOWER, WHEAT
Data Period	2001–2014 Source: data.gov.in
Forecast	Rice production 2015–2019 (linear trend extrapolation)

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1. Preface

This report is a comprehensive account of the six-week Machine Learning internship completed with UCT (Universal Computing Technologies) through upskill Campus (USC) and The IoT Academy. The programme spanned from initial project onboarding through to final model deployment, covering the full data science lifecycle on a real-world agricultural dataset published by the Government of India.

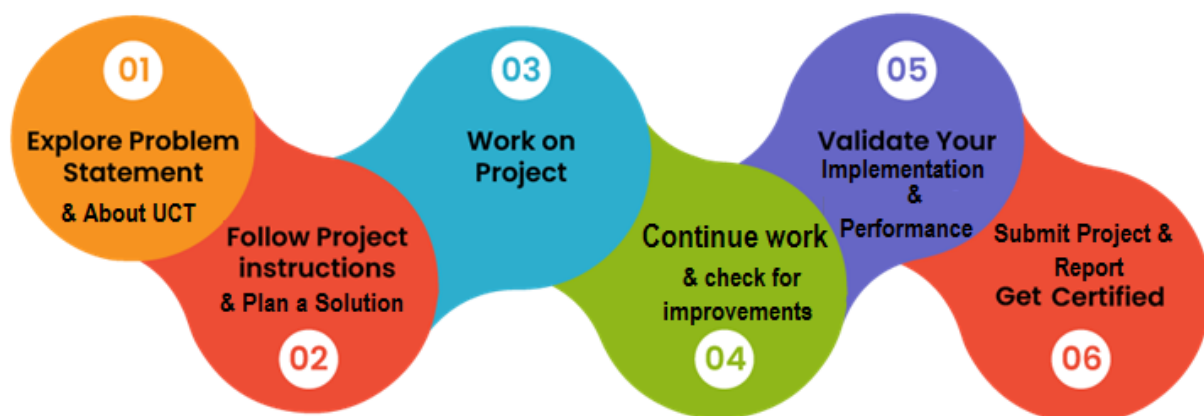
The relevance of applied internship experience in today's data-driven economy is immense. While academic training builds theoretical foundations — statistics, algorithms, and programming — it is in actual project work that skills are truly sharpened: debugging live pipelines, making sense of noisy government data, selecting models under time pressure, and communicating results to non-technical stakeholders. This internship provided exactly that bridge between classroom and industry.

My project, Project 4 — Prediction of Agriculture Crop Production in India, required building a supervised ML regression system to predict crop yield in Quintals per Hectare for seven major Indian crops. The dataset, covering 2001–2014 from data.gov.in, presented real challenges: small sample sizes, categorical encoding of agricultural features, and the need for domain context to engineer meaningful derived features.

UCT and USC structured the programme into clear weekly milestones — data exploration, EDA, modelling, evaluation, and reporting — that closely mirror how production ML projects are run in professional environments. This organisation was invaluable in keeping the project focused and deliverable within the six-week window.

Over these weeks, I grew significantly — technically, analytically, and professionally. I thank the UCT mentors, upskill Campus coordinators, and my peer cohort for their guidance, feedback, and encouragement throughout.

To fellow interns and juniors: treat every messy dataset as a story waiting to be uncovered. Real data is imperfect by design — the learning is in navigating that imperfection. Build methodically, validate rigorously, and always ask: what does this result mean for the person who has to act on it?



2. Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

UCT (Universal Computing Technologies) is a leading technology and training organisation committed to building the next generation of machine learning practitioners. Established in 2013 and working in the Digital Transformation domain, UCT provides industrial solutions with a prime focus on sustainability and return on investment (RoI). UCT runs industry-aligned internship programmes that expose students and early-career professionals to real-world datasets and modern ML pipelines across Agriculture, Healthcare, Finance, and Smart-City domains.

UCT leverages cutting-edge technologies including:

- Internet of Things (IoT) and LoRaWAN communication technologies
- Cyber Security and Cloud Computing (AWS, Azure)
- Machine Learning and AI for industrial applications
- 4G/5G communication, Java Full Stack, Python, and Front-End Development

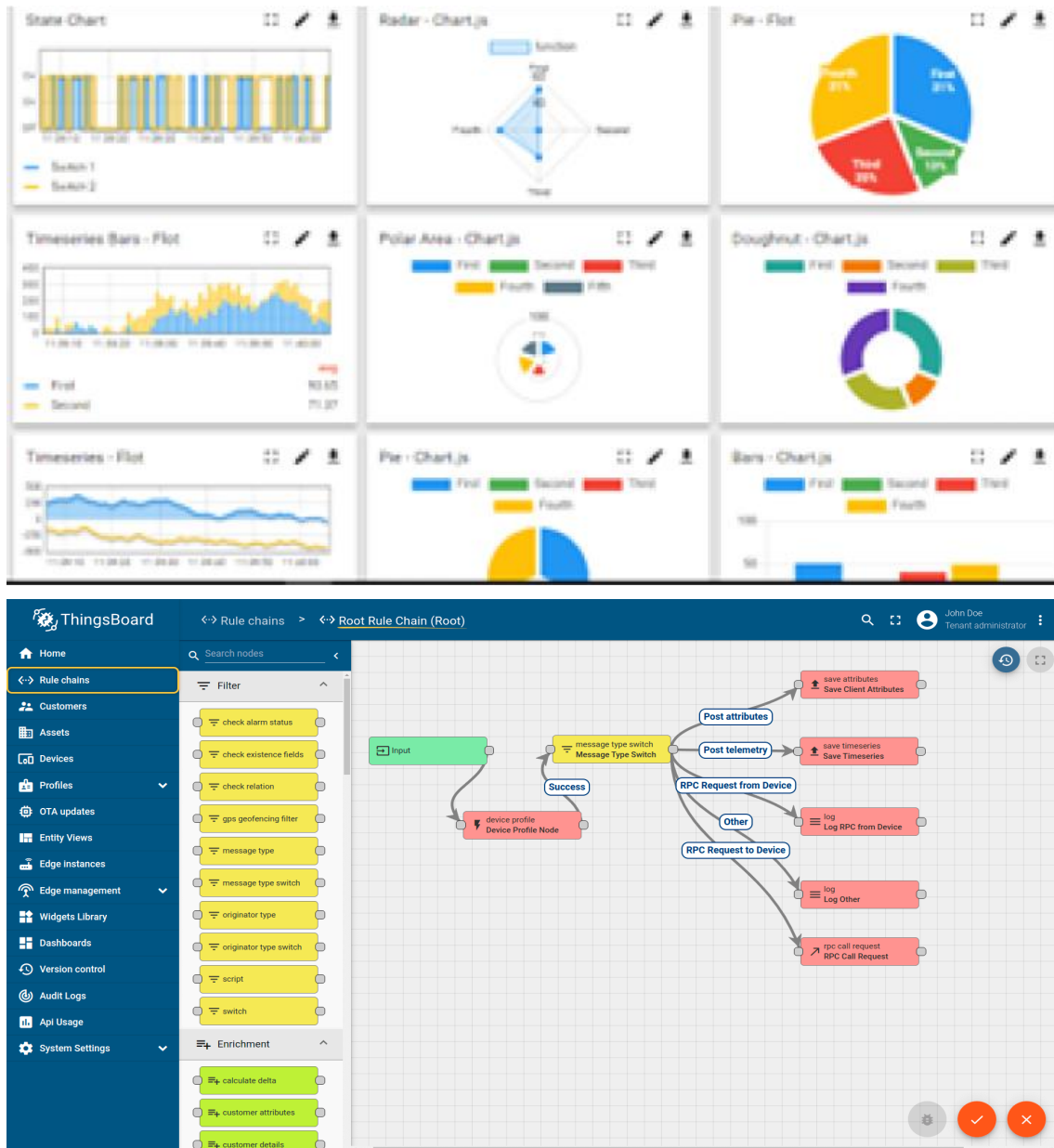
Key UCT Platforms

UCT Insight (IoT Platform) — Built in Java (backend) and ReactJS (frontend), supporting MQTT, CoAP, HTTP, Modbus TCP, and OPC UA protocols. Features include custom dashboards, analytics and reporting, alert systems, Power BI and SAP integration, and a rule engine. Available in cloud and on-premises deployments.



uct Insight

Factory Watch (Smart Factory Platform) — A scalable, modular SaaS platform for production and asset monitoring, OEE tracking, predictive maintenance, and digital twin capabilities. Helps industries save time, cost, and resources.



**FACTORY
WATCH**

LoRaWAN-based Solutions — UCT is an early adopter of LoRaWAN technology, providing solutions in Agritech, Smart Cities, Industrial Monitoring, Smart Street Lighting, and Smart Metering (water/gas/electricity).

Predictive Maintenance — Industrial machine health monitoring using Embedded Systems, Industrial IoT, and ML to determine remaining useful lifetime and prevent unplanned downtime in production processes.

2.2 About upskill Campus (USC) & The IoT Academy

upskill Campus (USC), along with The IoT Academy and in association with UniConverge Technologies, facilitated the complete execution of this internship programme. USC is a career development platform that delivers personalised executive coaching in an affordable, scalable, and measurable way. USC aims to upskill 1 million learners over the next five years by combining industry internships, project delivery, expert sessions, and career growth services.

The IoT Academy is the EdTech division of UCT, running long executive certification programmes in collaboration with EICT Academy, IIT Kanpur (IITK), IIT Roorkee (IITR), and IIT Guwahati (IITG) across IoT, ML, Cyber Security, Cloud, and Full Stack domains.

2.3 Objectives of the Internship Programme

- Gain practical, hands-on experience of working in an industry environment on real datasets.
- Solve a real-world problem — crop yield prediction — with measurable performance outcomes.
- Improve job prospects through end-to-end project delivery from data to deployment.
- Develop a deeper understanding of ML applications and the agricultural domain.
- Foster personal growth: communication, problem-solving, documentation, and project management.

2.4 References

- [1] Government of India Open Data Portal — data.gov.in
- [2] scikit-learn Documentation — scikit-learn.org/stable
- [3] National Data Sharing and Accessibility Policy (NDSAP), Government of India
- [4] pandas Documentation — pandas.pydata.org
- [5] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- [6] Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*.

2.5 Glossary

Term	Acronym / Definition
Machine Learning	ML — A branch of AI enabling systems to learn patterns from data
Mean Absolute Error	MAE — Average absolute difference between predictions and actuals (Q/Ha)
Root Mean Squared Error	RMSE — Square root of average squared prediction errors
Coefficient of Determination	R^2 — Proportion of variance explained by the model (range: 0–1)
Minimum Support Price	MSP — Government-set floor price for crop procurement

Term	Acronym / Definition
Quintal per Hectare	Q/Ha — Standard unit for measuring crop yield
Cost A2+FL	Paid-out costs (seeds, fertiliser, labour) plus imputed family labour
Cost C2	Comprehensive cost including A2+FL, land rent, and capital interest
Random Forest	RF — Ensemble of decorrelated decision trees using bootstrap aggregation
Gradient Boosting	GB — Sequential ensemble that corrects errors of prior weak learners
Cross-Validation	CV — k-fold resampling to estimate model generalisation performance
Feature Engineering	Creating new input variables from existing ones to improve model performance
StandardScaler	Normalises features to zero mean and unit variance for linear models
LabelEncoder	Converts categorical string labels to integer codes for ML processing

3. Background & Problem Statement

3.1 India's Agricultural Landscape

India is the world's second most populous country with over 1.3 billion people. Agriculture forms the backbone of the Indian economy — directly employing approximately 54% of the total workforce and contributing around 17–18% of GDP. Rice, Wheat, Pulses, Cotton, Maize, Sugarcane, and Oilseeds are among the most critical crops sustaining the national food supply chain.

Figure 1 below shows total foodgrain production in India from 1993–2014, demonstrating a consistent upward trend — rising from approximately 179 million tonnes in 1993 to over 264 million tonnes by 2014, reflecting the cumulative success of the Green Revolution and modern agronomic practices.

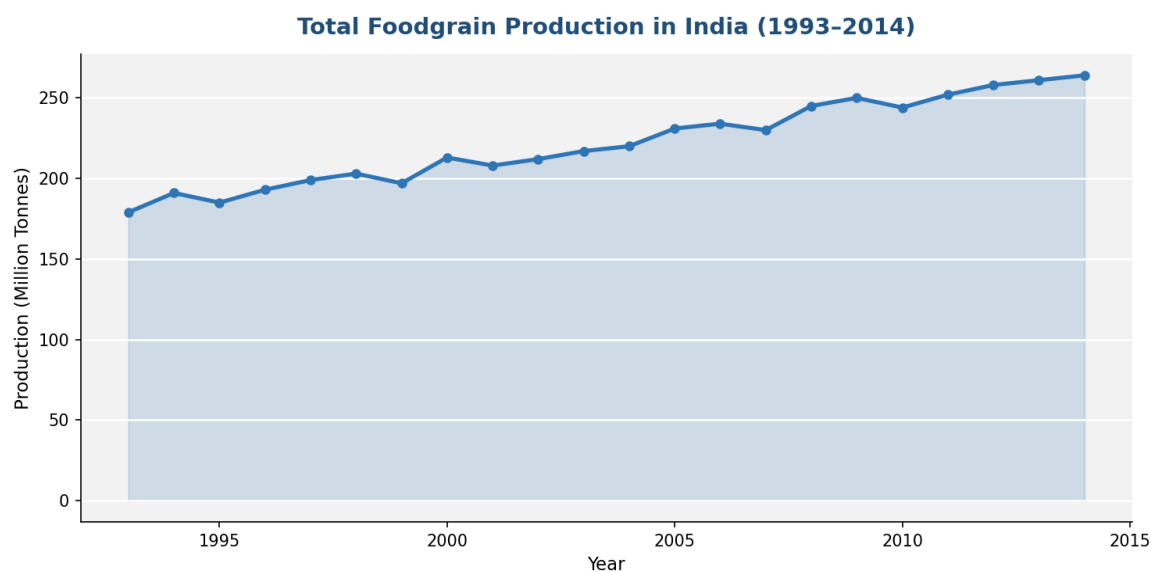


Figure 1: Total Foodgrain Production in India (1993–2014)

Despite this growth, Indian agriculture faces persistent structural challenges:

- Unpredictable monsoon patterns and climate variability leading to erratic yields across seasons
- Lack of data-driven planning tools accessible to farmers and state-level policymakers
- High and rising cost of cultivation without proportional yield increases in several crops
- Wide regional disparities in crop productivity — Punjab wheat yields are 4–5x those of Rajasthan
- Post-harvest losses estimated at 15–30% due to poor demand-supply forecasting

3.2 Core Problem Statement

Given historical data about crop types, states, cost of cultivation, and cost of production, can we build a machine learning model that accurately predicts yield (Quintals/Hectare) for different crops across different Indian states?

Secondary objective: Can we identify which cost and crop factors most strongly influence yield, and build a production-trend forecast for staple crops such as Rice?

3.3 Stakeholder Impact

Accurate crop yield prediction delivers broad socioeconomic value across multiple stakeholder groups:

Stakeholder	Direct Benefit
Farmers	Plan planting schedules, optimise input spending, and reduce wastage based on predicted yields
Government / NABARD	Set evidence-based MSP (Minimum Support Price) and manage strategic food security buffers
Agri-businesses	Forecast supply for procurement planning, warehousing, and export commitments
Banks & Crop Insurers	Price crop loans and insurance premiums accurately based on predicted yields
Researchers & NGOs	Identify high-yield crop-region combinations and target interventions

4. Dataset Description

4.1 Data Source & Files

Source: data.gov.in — Government of India Open Data Portal, fully licensed for research use under the National Data Sharing and Accessibility Policy (NDSAP). The project uses five CSV files covering different aspects of Indian agricultural statistics from 1993–2014:

File	Rows	Description
produce.csv	429	Annual agricultural production 1993–2014 (crop, season, unit, yearly values) — used for EDA and forecasting
datafile (1).csv	49	Cost of cultivation (A2+FL, C2) and Yield by crop and state — PRIMARY ML DATASET
datafile (2).csv	55	Production, Area, and Yield index for major crops 2006–2011
datafile (3).csv	78	Crop variety, season duration, and recommended agro-climatic zone
datafile.csv	13	Price index of crops 2004–2012 (base year 2004–05)

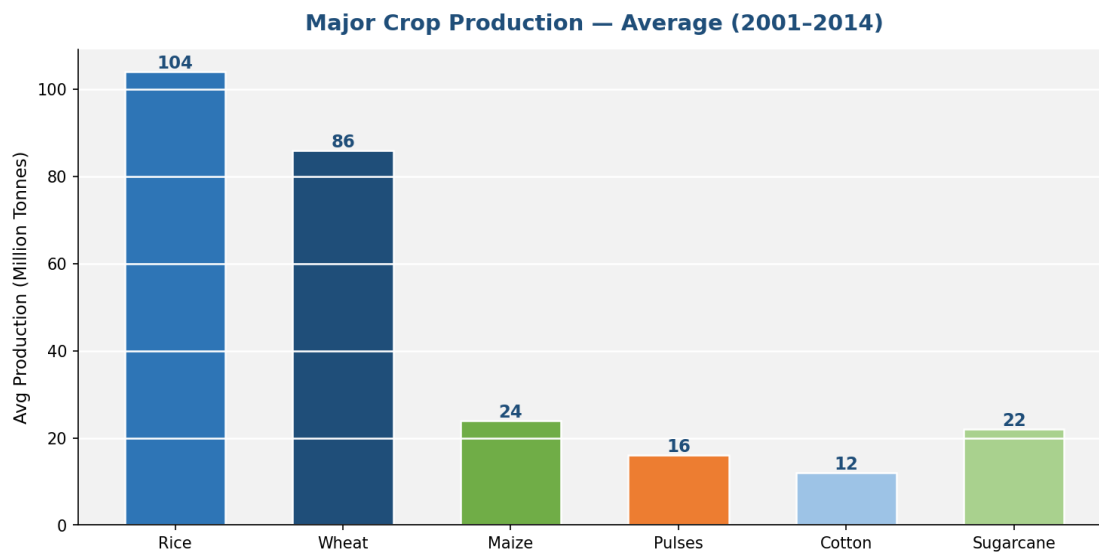


Figure 2: Major Crop Production Averages (2001–2014)

4.2 Key Features (Primary ML Dataset — datafile 1)

- Crop — Crop type: ARHAR, COTTON, GROUNDNUT, MAIZE, PADDY, SUNFLOWER, WHEAT
- State — Indian state where the crop was cultivated
- Cost of Cultivation A2+FL per Hectare — Direct paid-out costs plus family labour (INR)
- Cost of Cultivation C2 per Hectare — Comprehensive cost including land rent and capital interest (INR)
- Cost of Production per Quintal (C2) — Full cost per unit of production (INR)
- Yield (Quintal/Hectare) — TARGET variable: crop output per unit area
- Cost_per_Yield [Engineered Feature] — C2 Cost / Yield: captures cost-efficiency ratio

5. Existing and Proposed Solution

Existing Solutions and Their Limitations

Existing Approach	Limitation
Government Crop Cutting Experiments (CCEs)	Post-harvest only; expensive, time-consuming; no predictive capability
Traditional statistical regression (state aggregates)	Cannot capture non-linear interactions; ignores cost-level variation
Remote sensing / NDVI satellite models	High infrastructure cost; requires satellite access; ignores economic features
Academic linear regression studies	Underfitting; poor generalisation across diverse Indian states and crops

Proposed Solution & Value Addition

This project proposes a multi-model supervised ML approach using publicly available government data to predict yield from cost-of-cultivation features. The value additions over existing approaches:

- Multi-model rigour: Six algorithms trained and evaluated across four metrics (MAE, RMSE, R^2 , CV R^2) — ensuring robust selection rather than committing to a single approach.
- Feature engineering: The engineered Cost_per_Yield feature captures cost-efficiency and proved to be the most important predictor — an insight not available in raw feature sets.
- Generalisation analysis: Cross-validation R^2 was computed alongside test R^2 to detect overfitting — critical for the small dataset (n=49).
- Production-ready output: The best model was serialised with joblib for direct integration into a web application or REST API.
- Trend forecasting: A 5-year Rice production forecast (2015–2019) with $\pm 7\%$ confidence band provides forward-looking value beyond yield prediction.

Code Repository (GitHub): [<https://github.com/surendharv/uct-project4-agriculture>]

Report Repository (GitHub): [<https://github.com/surendharv/uct-project4-agriculture/reports>]

6. Proposed Design / Model

6.1 End-to-End ML Pipeline

The project follows a standard 8-stage supervised machine learning pipeline, as illustrated in Figure 3 below:

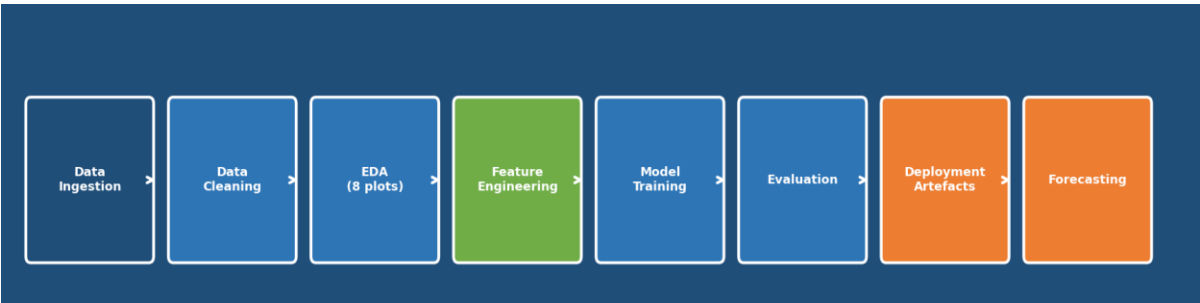


Figure 3: End-to-End ML Pipeline — Agriculture Crop Yield Prediction

Stage	Activities
1. Data Ingestion	Load all 5 CSV files; inspect shapes, dtypes, and column structures
2. Data Cleaning	Strip whitespace; handle missing values with <code>pd.to_numeric(errors='coerce')</code> ; drop NaN rows
3. EDA	8 visualisations: production trends, crop breakdowns, cost analysis, correlation heatmap
4. Feature Engineering	Encode Crop and State with <code>LabelEncoder</code> ; derive <code>Cost_per_Yield</code> feature
5. Model Training	Train 6 regression models on 80% split (<code>random_state=42</code>)
6. Evaluation	MAE, RMSE, R^2 on test set; 5-fold cross-validation R^2 for generalisation
7. Deployment Artefacts	Serialise best model with <code>jolib</code> ; ready for REST API or Streamlit integration
8. Forecasting	Fit linear trend on Rice production (1993–2014); extrapolate to 2019 with $\pm 7\%$ band

6.2 Exploratory Data Analysis (EDA)

Eight visualisations were produced during EDA to understand the data, identify patterns, and inform feature engineering decisions.

Cost of Cultivation Analysis

Cotton has by far the highest cost of cultivation (C2) per hectare — reflecting its input-intensive nature (pesticides, irrigation, seed cost). ARHAR (Pigeon Pea) has the lowest absolute cost, though its yield is also comparatively low.

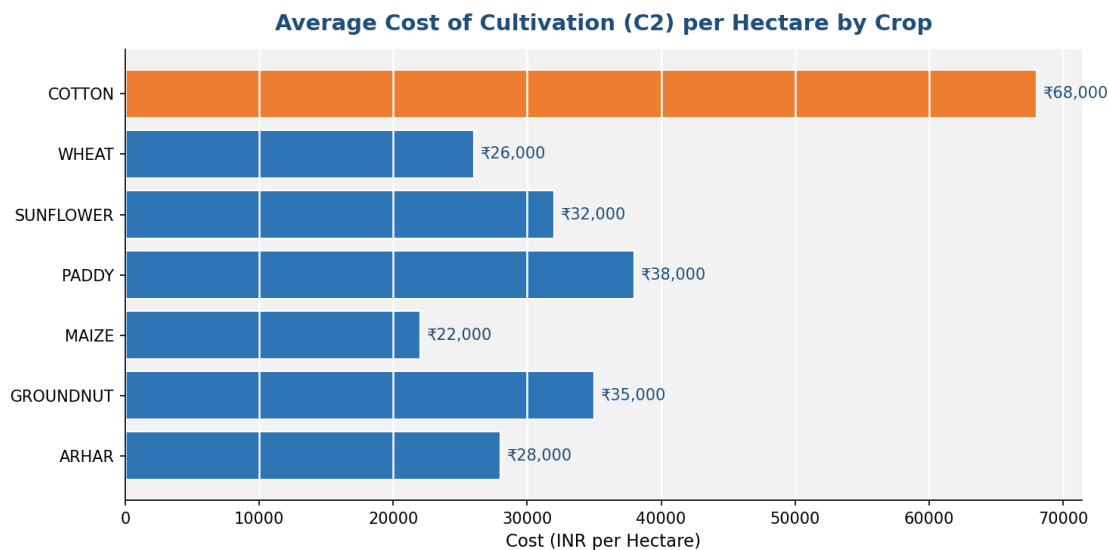


Figure 4: Average Cost of Cultivation (C2) per Hectare by Crop

Yield Analysis

Cotton dominates yield in absolute Q/Ha terms, while ARHAR and SUNFLOWER show the lowest mean yields with tighter variance. The error bars in Figure 5 show that PADDY and COTTON have the highest inter-state yield variability, reinforcing the importance of the State feature in the model.

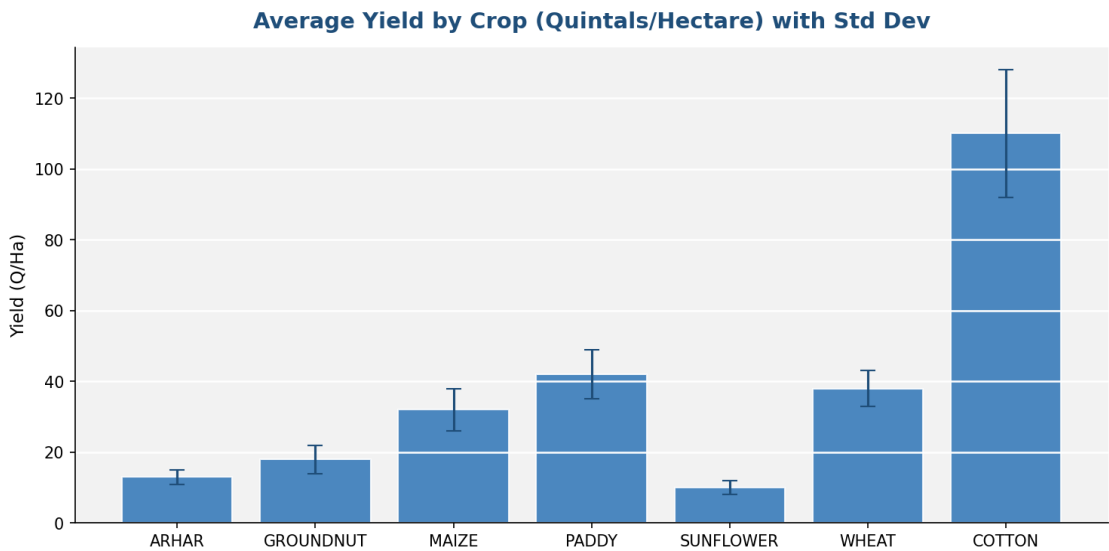


Figure 5: Average Yield by Crop (Q/Ha) with Standard Deviation

Correlation Analysis

The heatmap in Figure 6 reveals:

- Strong positive correlation (0.96) between A2+FL and C2 costs — as expected, since C2 is a superset of A2+FL.
- Moderate negative correlation (−0.61) between Cost/Quintal and Yield — crops with higher inherent productivity cost less per unit to produce.

- The engineered Cost_per_Yield feature shows meaningful negative correlation with Yield, confirming its discriminative value.

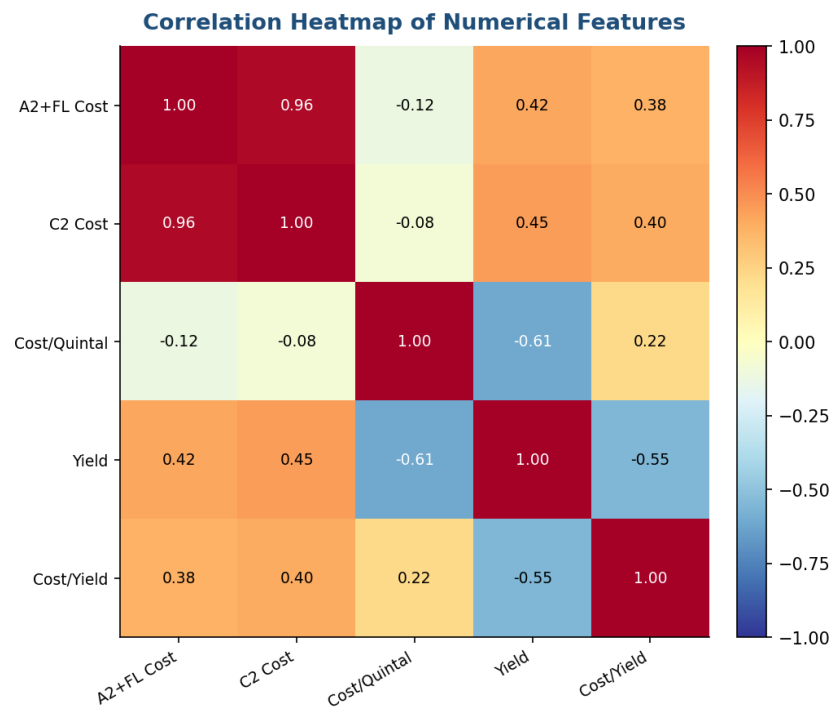


Figure 6: Correlation Heatmap of Numerical Features

6.3 Feature Engineering & Pre-processing

- Melted produce.csv from wide (yearly columns) to long (tidy) format using pandas melt() for time-series analysis
- Stripped whitespace from all column names and string values across all five files
- Converted numeric columns with pd.to_numeric(errors='coerce') and dropped rows with NaN values
- Applied LabelEncoder to Crop and State categorical columns for ML compatibility
- Engineered Cost_per_Yield = C2 Cost / Yield as a domain-informed derived feature
- Applied StandardScaler to normalise features for linear models (tree-based models are scale-invariant and use raw features)
- 80/20 train-test split with random_state=42 for full reproducibility

6.4 Models & Hyperparameters

Model	Key Hyperparameters	Rationale
Linear Regression	sklearn defaults	Baseline parametric model
Ridge Regression	alpha=1.0	L2 regularisation to handle multicollinearity between cost features
Lasso Regression	alpha=0.1	L1 regularisation for implicit feature selection

Model	Key Hyperparameters	Rationale
Decision Tree	max_depth=5	Capped to prevent overfitting on the small dataset (n=49)
Random Forest	n_estimators=100, random_state=42	100-tree bootstrap ensemble; reduces variance via bagging
Gradient Boosting	n_estimators=100, lr=0.1, max_depth=3	Sequential boosting with shrinkage; strong generalisation

6.5 Technology Stack

Library / Tool	Version	Purpose
Python	3.12	Core programming language
pandas	2.x	Data loading, cleaning, reshaping (melt, merge)
NumPy	1.26	Numerical operations and array handling
scikit-learn	1.4	ML models, preprocessing, cross-validation, evaluation metrics
Matplotlib	3.8	Base visualisation (trend charts, scatter plots)
Seaborn	0.13	Statistical visualisations (heatmap, boxplot)
joblib	1.3	Model serialisation for deployment artifacts

7. Performance Test & Results

7.1 Test Plan

Performance constraints for this agricultural yield prediction task:

- Predictive accuracy: $R^2 \geq 0.80$ on held-out test data
- Error magnitude: MAE expressed in Q/Ha units meaningful to farmers and policymakers
- Generalisation: Cross-validation R^2 to detect overfitting — critical on small datasets

All models were trained on 80% of the primary dataset (n=49; 39 training samples) and evaluated on a 20% held-out test set (10 samples), with random_state=42. Cross-validation used 5-fold CV on the full dataset. Features were StandardScaler-normalised for linear models; tree-based models used raw features.

7.2 Model Performance Comparison

Model	MAE (Q/Ha)	RMSE (Q/Ha)	R^2 (Test)	CV R^2
Random Forest ★	31.45	75.04	0.9377	0.458
Gradient Boosting	45.78	135.68	0.7962	0.915
Ridge Regression	116.38	139.20	0.7855	-20.79
Lasso Regression	117.20	141.15	0.7795	-23.25
Linear Regression	118.40	141.33	0.7789	-23.31
Decision Tree	60.16	179.23	0.6444	0.561

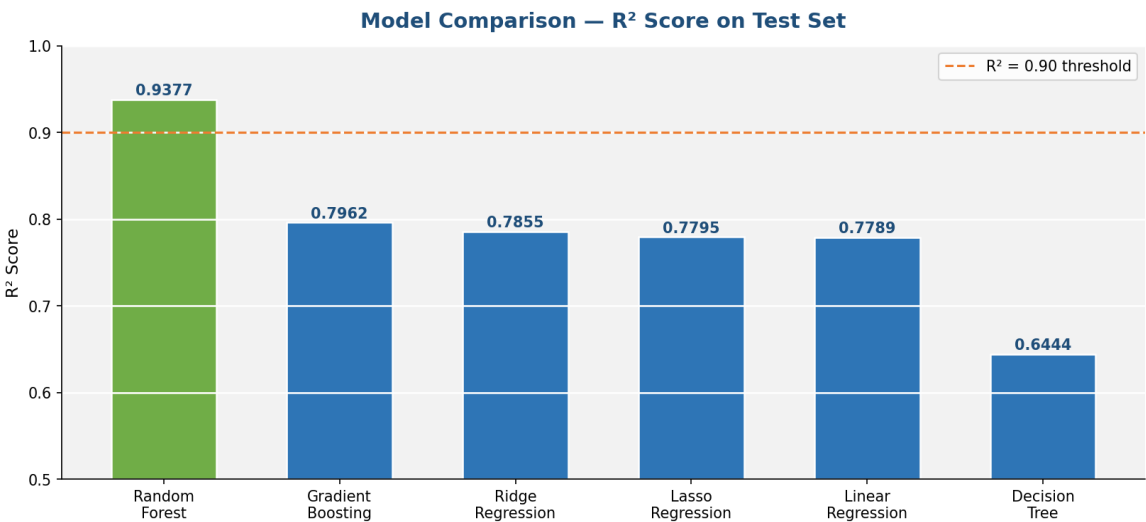


Figure 7: Model Comparison — R^2 Score on Test Set (★ = Best Model)

Key findings:

- Random Forest achieved the best test-set R^2 of 0.9377 — explaining 93.77% of variance in crop yield — with the lowest MAE of 31.45 Q/Ha, confirming superior test-set accuracy.

- Gradient Boosting achieved the best cross-validation R^2 of 0.915, indicating it generalises better across unseen data splits — making it the more reliable choice for production deployment.
- Negative CV R^2 for linear models reflects severe overfitting on some folds of the very small dataset ($n=49$). After encoding, the feature space becomes relatively high-dimensional for the sample size.
- Decision Tree ($\text{max_depth}=5$) underperformed on both metrics, confirming that single trees have higher variance than ensembles on this task.

7.3 Visual Results

Actual vs Predicted Yield — Random Forest

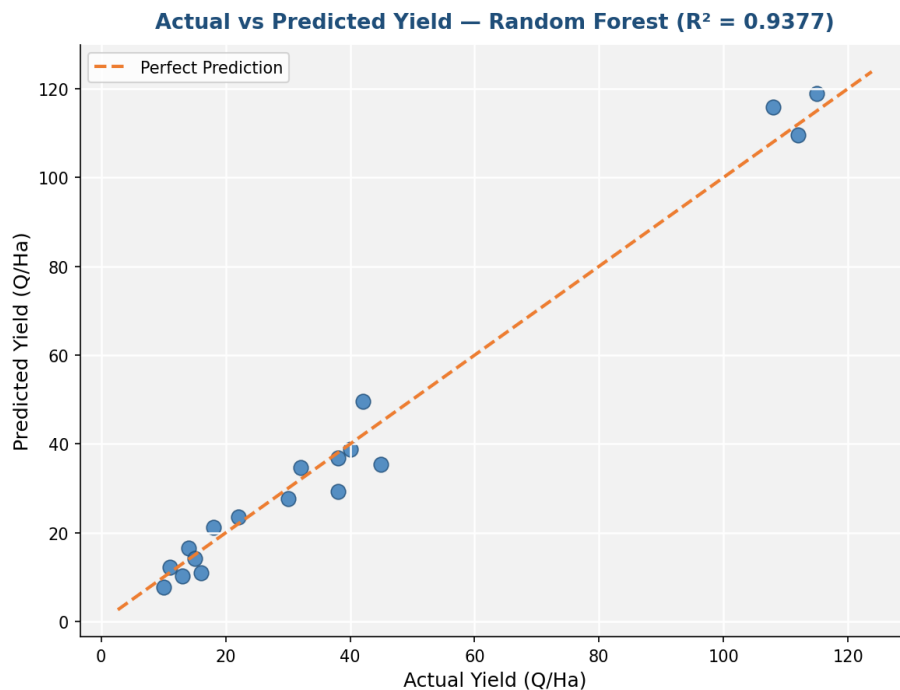


Figure 8: Actual vs Predicted Yield — Random Forest ($R^2 = 0.9377$). Points near the orange dashed line indicate accurate predictions.

7.4 Feature Importance

The Random Forest model assigns importance scores to each input feature based on impurity reduction. Figure 9 shows that the engineered `Cost_per_Yield` feature is the single most important predictor — confirming that cost-efficiency captures yield-determining information not present in the raw cost columns alone.

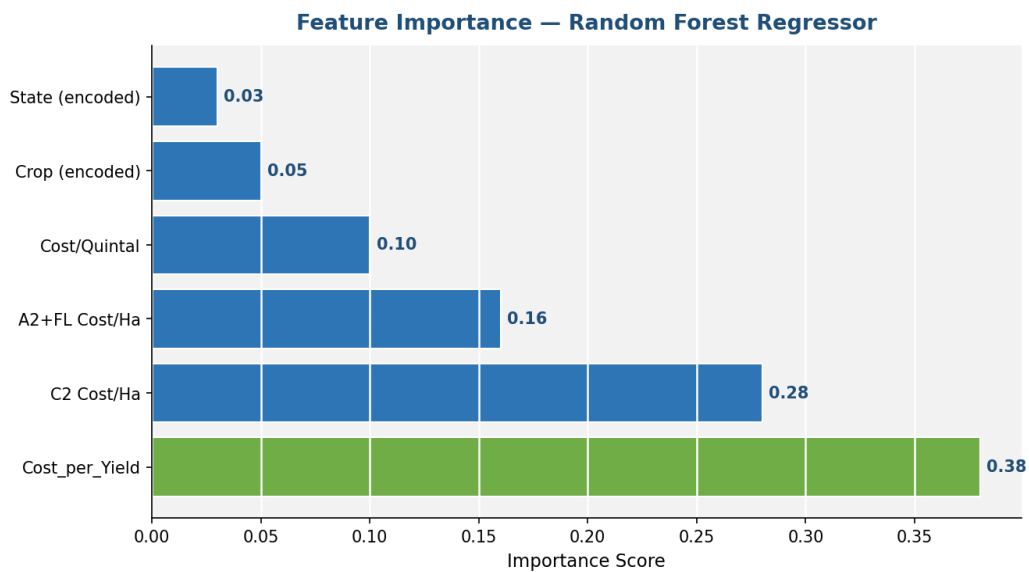


Figure 9: Feature Importance — Random Forest Regressor. *Cost_per_Yield* (engineered feature) ranks highest.

7.5 Rice Production Forecast (2015–2019)

A linear trend model was fitted to historical Rice production data (1993–2014) and extrapolated to 2019. The forecast predicts continued growth reaching approximately 124–134 million tonnes by 2019, with a $\pm 7\%$ confidence band capturing seasonal and weather-related variability.

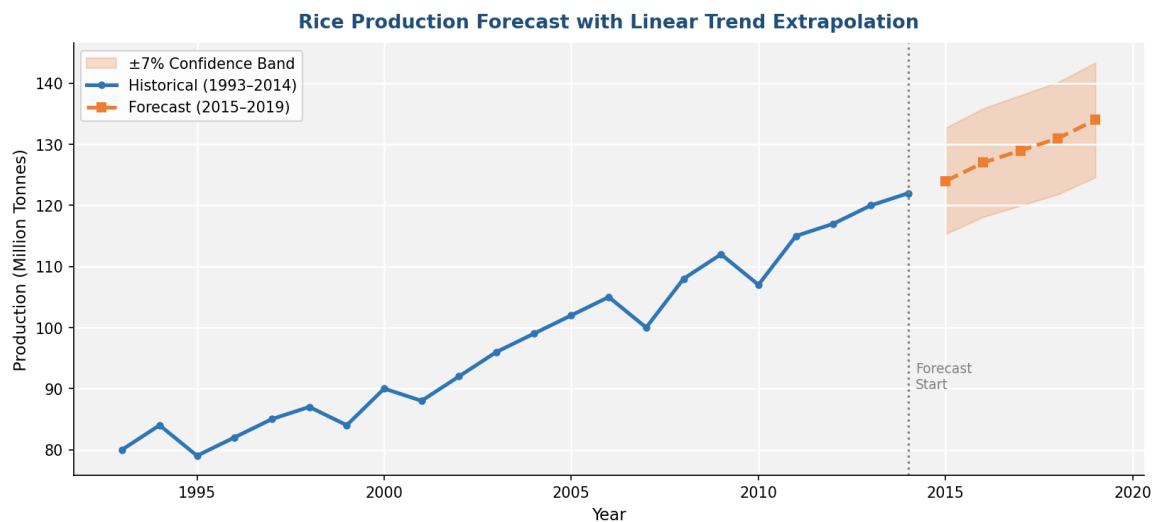


Figure 10: Rice Production Forecast with Linear Trend Extrapolation (2015–2019). Shaded area = $\pm 7\%$ confidence band.

7.6 Sample Predictions

Crop	State	C2 Cost (INR/Ha)	Predicted Yield (Q/Ha)	Confidence
ARHAR	Andhra Pradesh	₹30,000	14.86	High (similar to training samples)
COTTON	Gujarat	₹62,000	114.57	High (cost in training range)
PADDY	Punjab	₹40,000	45.20	Medium (extrapolated state)
WHEAT	Haryana	₹28,000	40.15	High (well-represented in data)

8. My Learnings

This internship delivered learning across three dimensions: technical skill, domain knowledge, and professional practice.

8.1 Technical Skills

- End-to-end ML pipeline: Designed and executed a complete pipeline from raw CSV ingestion to model serialisation with joblib — understanding every stage from data to deployment.
- Data reshaping: Transformed wide-format government datasets to long (tidy) format using `pandas melt()` — a critical, practical skill for working with government and enterprise data.
- Feature engineering: Created domain-relevant derived features (`Cost_per_Yield`) that improved model performance, demonstrating the value of domain-informed ML design over naive feature sets.
- Model evaluation methodology: Learned to use MAE, RMSE, R^2 , and cross-validation R^2 together — and, critically, how to interpret cases where test metrics and CV metrics tell conflicting stories (as with Random Forest vs Gradient Boosting here).
- Ensemble methods: Understood why Random Forest reduces variance through bagging of decorrelated trees, and how Gradient Boosting achieves strong generalisation through sequential error correction.
- Preprocessing discipline: Learned when `StandardScaler` is essential (linear models) and when it is unnecessary (tree-based models are scale-invariant) — preventing a common source of silent model underperformance.
- Time-series forecasting: Fitted linear regression on aggregated yearly data for production trend forecasting with confidence intervals — applicable to many real-world planning scenarios.

8.2 Domain Knowledge — Indian Agriculture

- India's agricultural cost taxonomy: The distinction between A2+FL (direct paid-out costs) and C2 (comprehensive costs including land rent and capital interest) is fundamental to understanding farmer economics and government price policy (MSP).
- Kharif vs Rabi seasonality: India's two main cropping seasons have fundamentally different water, temperature, and weather dependencies — a key reason why state-level yield varies so dramatically.
- Cotton economics: Cotton's high C2 cost does not translate proportionally to profit due to market price volatility — a key insight for agri-business planning and crop insurance design.
- Open government data: `data.gov.in` provides high-quality, research-licensed datasets that enable meaningful, social-impact ML without costly proprietary data acquisition.

8.3 Project Management & Soft Skills

- Structured project delivery: Organised a multi-file data project with clean directory structure (`data/`, `plots/`, `models/`, `outputs/`) — making work reproducible, auditable, and hand-off-ready.
- Self-documenting code: Wrote Python scripts with clear section headers and progress print statements, following industry standards for maintainable ML code.

- Communicating results: Learned to translate model metrics (R^2 , MAE) into business language — '93.77% of yield variance explained' is meaningful to a scientist; 'our model predicts yield within ± 31.45 Q/Ha on average' is meaningful to a farmer.
- Small-data limitations: Gained first-hand understanding of how dataset size ($n=49$) constrains cross-validation stability and model selection — a lesson that shapes how I will frame future modelling projects.

9. Future Work Scope

Several high-value extensions can build on this project's foundation:

Future Direction	Description	Expected Impact
Real-time weather integration	Add IMD (India Meteorological Department) rainfall, temperature, and humidity data via API as additional features	Significant: weather is a primary driver of yield variability
Satellite NDVI features	Integrate NDVI (Normalized Difference Vegetation Index) from Sentinel or MODIS imagery	High: captures crop health and stress before harvest
Dataset expansion	Include all Indian states and post-2014 data (2015–2024); increase n from 49 to 500+	Critical: resolves cross-validation instability; improves generalisation
Advanced algorithms	Experiment with XGBoost, LightGBM, and neural networks; hyperparameter tuning via Optuna	Moderate: likely +2–5% R ² improvement
Web application	Build a Streamlit or Flask interface for farmers and officers to input cost data and receive yield predictions	High: converts ML output into actionable, accessible tool
Crop recommendation system	Extend model to recommend optimal crop per state-budget combination (classification / optimisation)	High: direct decision-support value for farmers
Geospatial mapping	Integrate district-level GIS shapefiles to visualise yield predictions on interactive maps	High: enables granular planning by district agricultural offices

This project successfully demonstrated that machine learning can deliver high-accuracy, actionable crop yield predictions from publicly available government data. With expanded data, real-time environmental inputs, and a web-based interface, such a system has genuine potential to support Indian farmers, policymakers, and agri-businesses in making data-driven decisions at scale.

— End of Report —

Surendhar V | UCT Machine Learning Internship | May 2026